

Teaser figure pending completed runs

Figure 1: **SHIFTWM forecasts by moving observed features.** Given the last observed frame of real robot and surgical video and a sequence of future actions, SHIFTWM predicts, for every future step, where each patch’s features come from (arrows: learned transport field), where new content must be synthesised (gate), and a correction. Recursive predictors regenerate the full feature grid at every step and drift (right).

SHIFTWM: ACTION-CONDITIONED TRANSPORT OF OBSERVED FEATURES FOR VISUAL WORLD MODELS

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ABSTRACT

World models that forecast in the feature space of a frozen visual foundation model are fast enough for planning, but they must regenerate every future patch from scratch, even though most of a manipulation scene only moves. We introduce SHIFTWM, a latent world model that forecasts by *moving what it has already seen*. For each prefix of future actions, SHIFTWM predicts a local soft transport of the observed patch features, a gate that decides where transport is used, and a learned correction for content that transport cannot explain. Each forecast is computed directly from measured features rather than from the model’s own earlier predictions, so errors do not compound through recursion, and the transport field is a readable estimate of predicted motion. We evaluate on held-out real robot video (DROID), on surgical robot video with recorded kinematics (OpenH-Embodiment, seven dVRK tasks), on single-image manipulation forecasting (IWS), and on standard planning benchmarks (PushT, TwoRoom, Reacher). All comparisons use matched backbones and frozen encoders (DINOv2, DINOv3). We also measure action sensitivity, a property that anchored predictors can lose, and show how an action-contrastive objective recovers it for planning.

1 INTRODUCTION

A world model predicts how observations change in response to actions. Recent work shows that forecasting in the feature space of a frozen, self-supervised encoder such as DINOv2 or V-JEPA 2 gives compact models that support planning without generating pixels (Zhou et al., 2025; Assran et al., 2025; Maes et al., 2026b; Sobal et al., 2025). These predictors are *generative in feature space*. At every future step they output the complete grid of patch features, usually by feeding their own predictions back as inputs. In manipulation and surgical video, however, most of the scene persists:

tissue, tools, the table and the robot move, but they are rarely created or destroyed. Regenerating persistent content at every step wastes capacity and lets errors compound under recursion.

We ask a simple question. *Can a latent world model forecast by transporting features it has already observed, and synthesise only what transport cannot explain?* Motion-based video prediction answered a version of this question in pixel space (Finn et al., 2016; Ebert et al., 2017; De Brabandere et al., 2016; Liu et al., 2017). The idea has not been tested for foundation-feature world models used for planning, where features are semantic, high-dimensional and coarse in space. Three questions are open: whether transport helps once the predictor is strong, whether it survives held-out real data, and what it costs for control.

We introduce SHIFTWM (Figures 1 and 2). A memory encoder reads the patch features of the last few observed frames. For each future step, a decoder conditioned on the causal action-prefix state produces per-patch queries. Each query (i) attends to a local window of *observed* feature locations and copies their content, (ii) gates between staying in place and moving, and (iii) adds a learned correction. Every horizon is predicted directly from measured features, so errors do not compound through recursion, and the transport weights form a motion field that we can inspect and compare against optical flow.

Contributions.

- **Method.** SHIFTWM, an action-conditioned local feature transport decoder for latent world models. It is a drop-in output head for any frozen patch encoder and adds $< 2\%$ parameters.
- **Evidence across domains.** Matched comparisons against recursive (DINO-WM-style), direct-additive and cross-attention predictors on held-out real robot video (DROID), seven surgical dVRK tasks (Open-H-Embodiment), IWS manipulation forecasting and three planning benchmarks. We use two encoder families and report forecasting, decoded-pixel, probing, action-sensitivity and planning metrics.
- **Analysis.** We show that the learned transport field tracks measured motion. We also characterise a trade-off specific to anchored predictors: open-loop accuracy against action sensitivity. An action-contrastive objective resolves this trade-off for planning.

2 RELATED WORK

World models in foundation-feature and JEPA latent spaces. A recent line of work predicts dynamics in the embedding space of a pretrained or jointly learned encoder instead of in pixels. DINO-WM trains a causal ViT predictor on frozen DINOv2 patch features and plans with model-predictive control (Zhou et al., 2025; Oquab et al., 2024); PLDM, LeWM and JEPA-WMs study joint-embedding predictive models learned from offline trajectories (Sobal et al., 2025; Maes et al., 2026b; Balestriero & LeCun, 2025; Terver et al., 2026), and V-JEPA 2-AC post-trains an action-conditioned predictor on DROID for zero-shot robot planning (Assran et al., 2025; Khazatsky et al., 2024). DINO-world, DINO-Foresight, VFMM and FlowWM forecast future foundation features, deterministically or generatively, and evaluate them through downstream dense heads (Baldassarre et al., 2025; Karypidis et al., 2025; Boduljak et al., 2025; Porcher et al., 2026). DDP-WM separates sparse dynamic regions from a static background, and RLA-WM models residual latent actions over visual features (Yin et al., 2026; Zhang et al., 2026a). Complementary analyses ask which latent spaces and objectives make predictions action-relevant (Nilaksh et al., 2026; Yeom et al., 2026; Zhang et al., 2026b; Huang et al., 2026). In these models the predictor synthesizes each future feature map and, in most cases, conditions later steps on its own outputs. ShiftWM keeps the same frozen-encoder interface, applicable to DINOv2 or DINOv3 features (Siméoni et al., 2025), but changes the output parameterization: every forecast is expressed as a transport of observed features plus a correction, so the predictor is never required to regenerate static content.

Generative world models and model-based reinforcement learning. Pixel-space generative world models learn high-fidelity, action-controllable video simulators. NWM conditions a diffusion transformer on navigation actions (Bar et al., 2025); Cosmos and Cosmos-Predict2.5 provide large video foundation models for physical AI (NVIDIA, 2025a;b) that Cosmos Policy adapts for control (Kim et al., 2026); Ctrl-World, IRASim, UniSim and IWS target robot manipulation and policy evaluation (Guo et al., 2026b; Zhu et al., 2025; Yang et al., 2024; Wang et al., 2026); and the Genie

family generates interactive environments from video, image or text prompts (Bruce et al., 2024; Parker-Holder & Fruchter, 2025). Model-based reinforcement learning instead learns task-specific latent dynamics jointly with rewards and policies, from PlaNet and DreamerV3 to TD-MPC2, IRIS and DIAMOND (Hafner et al., 2019; 2025; Hansen et al., 2024; Micheli et al., 2023; Alonso et al., 2024). These systems optimize sample fidelity or online return with their own encoders, data and, often, billions of parameters, so they do not share our contract of a fixed frozen encoder, identical observations and actions, and targets in common feature coordinates. We therefore treat them as complementary rather than as compute-matched baselines: they address rendering fidelity and online control, whereas we isolate the output parameterization of a compact predictor on a fixed representation.

Transformation- and transport-based prediction. Predicting the future by moving existing content has a long history in video prediction. CDNA and DNA predict action-conditioned convolutional kernels or pixel-motion distributions that transform the previous frame (Finn et al., 2016), dynamic filter networks generate input-dependent filters (De Brabandere et al., 2016), and Vondrick and Torralba transform past pixels with adversarially trained transformers (Vondrick & Torralba, 2017). SNA adds temporal skip connections to the first observed image to handle occlusion (Ebert et al., 2017), while flow-warping approaches synthesize frames by resampling along predicted voxel flow (Liu et al., 2017); SAVP and MCVD extend pixel prediction with stochastic and diffusion models (Lee et al., 2018; Voleti et al., 2022). Recent robot world models reintroduce explicit motion: FlowDreamer predicts 3D scene flow before diffusion rendering (Guo et al., 2026a), AHEAD forecasts VLA patch tokens conditioned on optical-flow velocity (Syed et al., 2026), and anchored navigation world models resist drift through anchor frames and epipolar guidance (Luan et al., 2026). Object-centric models such as SlotFormer and SOLD impose structure through slots (Wu et al., 2023; Mosbach et al., 2025). ShiftWM differs from these designs in four respects. It transports frozen foundation features rather than pixels; its source is a fixed memory of *observed* feature grids that remains unchanged at every horizon, so predicted states are never fed back as inputs; each horizon- h forecast is produced directly from the action prefix $a_{1:h}$; and the transport is a local softmax over a spatial neighborhood, combined with a gated identity path and a learned correction for disocclusion and appearance change.

World models for surgery and evaluation suites. Surgical world modeling is driven by generative video models: Endora simulates endoscopy video (Li et al., 2024), SurgWM learns latent-action-controllable surgical video (Koju et al., 2025), SurgVista targets long-horizon instrument-tissue dynamics (Pan et al., 2026), SWoMo combines neural and symbolic components for cataract surgery (Sivakumar et al., 2026), and Cosmos-H-Surgical learns surgical robot policies from videos through world modeling (He et al., 2025). Open-H-Embodiment releases paired video and kinematics across surgical platforms, together with a kinematics-conditioned surgical simulator (Open-H-Embodiment Consortium, 2026); SutureBot, JIGSAWS, Cholec80 and CholecT50 provide surgical demonstrations and task, skill and workflow annotations (Haworth et al., 2025; Gao et al., 2014; Twinanda et al., 2017; Nwoye et al., 2022), and SurgWMBench benchmarks instrument-motion planning (Liu et al., 2026). We use Open-H dVRK recordings to test kinematics-conditioned forecasting in feature space, a regime these pixel generators do not target. More broadly, robot datasets such as DROID, BridgeData V2 and Open X-Embodiment supply action-labeled video (Khazatsky et al., 2024; Walke et al., 2023; Open X-Embodiment Collaboration, 2024). WorldScore, World-ModelBench, EWMBench and Physics-IQ score generated video for controllability and physical plausibility (Duan et al., 2025; Li et al., 2025; Yue et al., 2025; Motamed et al., 2026), whereas stable-worldmodel standardizes latent planning evaluation (Maes et al., 2026a). We follow the latter protocols for control and report feature-space forecasting error for prediction.

3 METHOD

Problem. A frozen encoder E maps an image to a grid of $N = G \times G$ patch features $\mathbf{Z}_t \in \mathbb{R}^{N \times C}$. Given H observed grids $\mathbf{Z}_{-H+1:0}$, the actions between them $\mathbf{a}_{-H+1:-1}$, and a future action sequence $\mathbf{a}_{0:K-1}$, a world model predicts $\hat{\mathbf{Z}}_{1:K}$. The forecast $\hat{\mathbf{Z}}_k$ may depend only on $\mathbf{a}_{<k}$ (causality). Features are standardised per channel with train-split statistics that are shared across positions.

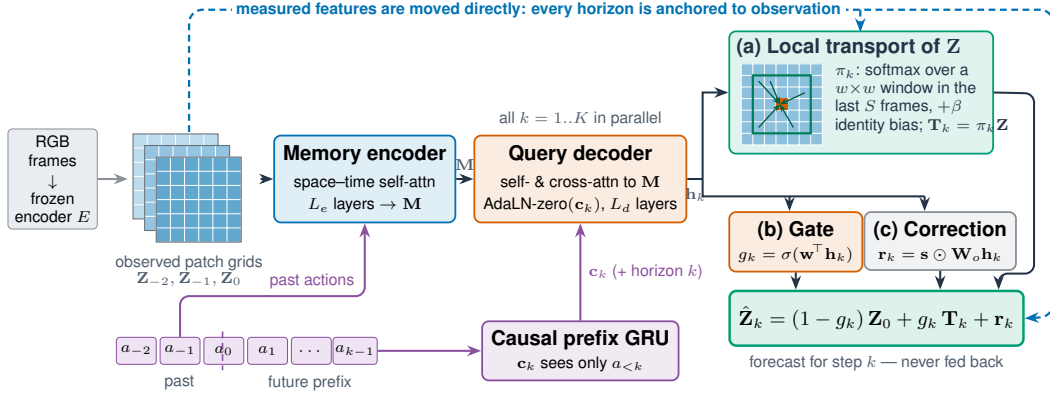


Figure 2: **SHIFTWMM architecture.** A frozen encoder maps the H observed frames to patch grids $\mathbf{Z}_{-H+1:0}$. The memory encoder mixes them over space and time. For each future step k , a decoder conditioned on the causal action-prefix state $\mathbf{c}_k$ produces per-patch hidden states $\mathbf{h}_k$. The output combines three terms: (a) local soft transport of *observed* features, (b) a gate between the identity path and transport, and (c) a scaled correction. Every horizon reads measured features directly, so no forecast is fed back as input.

Memory encoder. Each observed patch feature is projected to width d and receives a spatial embedding, a frame embedding and an embedding of the action applied after that frame. L_e transformer layers over all $H \cdot N$ tokens give memory states $\mathbf{M} \in \mathbb{R}^{H \times N \times d}$.

Action-prefix queries. A unidirectional GRU over embedded actions gives a causal prefix state. We add a horizon embedding to form $\mathbf{c}_k$. Query tokens start from the memory states of the last frame, and L_d decoder layers apply self-attention across patches and cross-attention to $\mathbf{M}$. Each layer is modulated by $\mathbf{c}_k$ through adaptive layer normalisation with zero-initialised gains (Peebles & Xie, 2023). The result is $\mathbf{h}_{k,i} \in \mathbb{R}^d$ for patch i at step k . All K steps are decoded in parallel.

Local feature transport. For query patch i , let $\mathcal{W}(i)$ be the $w \times w$ spatial window around i in each of the last $S \leq H$ observed frames. Transport weights are

$$\pi_{k,i}(j) = \text{softmax}_{j \in \mathcal{W}(i)} \left(\frac{1}{\sqrt{d_k}} (\mathbf{W}_q \mathbf{h}_{k,i})^\top (\mathbf{W}_k \mathbf{M}_j) + \beta \mathbb{I}[j = (0, i)] \right), \quad (1)$$

where β is a learned identity bias toward the same location in the last frame. Transport moves *measured* features, not latent states: $\mathbf{T}_{k,i} = \sum_j \pi_{k,i}(j) \mathbf{Z}_j$.

Gated output. A per-patch gate $g_{k,i} = \sigma(\mathbf{w}_g^\top \mathbf{h}_{k,i} + b_g)$ and a correction $\mathbf{r}_{k,i} = \mathbf{s} \odot \mathbf{W}_o \mathbf{h}_{k,i}$ give

$$\hat{\mathbf{Z}}_{k,i} = (1 - g_{k,i}) \mathbf{Z}_{0,i} + g_{k,i} \mathbf{T}_{k,i} + \mathbf{r}_{k,i}. \quad (2)$$

$\mathbf{W}_o$ is zero-initialised and $b_g = -3$, so an untrained model starts close to persistence and learns departures from it. $\mathbf{s} \in \mathbb{R}^C$ is a learned per-channel scale. We found that an earlier bounded (tanh) correction hurts on held-out data (Appendix D).

Training. We minimise the mean squared error over all horizons, $\mathcal{L}_{\text{pred}} = \frac{1}{K} \sum_k \|\hat{\mathbf{Z}}_k - \mathbf{Z}_k\|^2$. Anchoring every forecast to observed features makes it easy to ignore actions: a forecast that stays close to $\mathbf{Z}_0$ scores well on average. For planning we therefore add an action-contrastive hinge. For each sample it compares the prediction error under the true future actions with the error under actions borrowed from another sample in the batch:

$$\mathcal{L}_{\text{act}} = \max(0, m - (\|\hat{\mathbf{Z}}(\tilde{\mathbf{a}}) - \mathbf{Z}\|^2 - \text{sg} \|\hat{\mathbf{Z}}(\mathbf{a}) - \mathbf{Z}\|^2)), \quad \mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{act}}. \quad (3)$$

Matched baselines. To isolate the output head, all learned predictors share the same memory encoder, decoder, width and depth. **Direct** replaces Equation (2) with $\mathbf{Z}_0 + \mathbf{W}_o \mathbf{h}_k$. It has the

Table 1: **Held-out forecasting.** Standardised feature MSE ($\downarrow$) averaged over horizons 1–10, on test splits never used for selection. DINOv2-S patch features, three seeds.

Method	DROID (cam 1)	DROID (cam 2)	Open-H Hamlyn	IWS
Persistence	pending	pending	pending	pending
Linear extrapolation	pending	pending	pending	pending
AR-TF (DINO-WM-style)	pending	pending	pending	pending
AR (rollout-trained)	pending	pending	pending	pending
Direct (cross-attention)	pending	pending	pending	pending
SHIFTWM (ours)	pending	pending	pending	pending

same cross-patch attention, so it controls for spatial communication. **AR** predicts a residual to the most recent grid and is rolled out recursively, with gradients through the rollout. **AR-TF** follows DINO-WM: it is trained with teacher forcing to predict absolute features and is rolled out at test time. **Persistence** copies $\mathbf{Z}_0$, and **Linear** extrapolates the last feature difference.

Planning. For control we plug each predictor into the same model-predictive planner (CEM) with the official budgets of each benchmark. The cost is the feature distance between the predicted grid and the goal grid (Section 4.4).

4 EXPERIMENTS

We ask four questions. **(Q1)** Does transport improve held-out forecasting over matched predictors on real robot and surgical video? **(Q2)** Does the gain hold across encoders and horizons? **(Q3)** What happens in closed-loop planning? **(Q4)** What does the transport field learn? All numbers come from completed runs with three training seeds. Model and checkpoint selection use validation splits only, and test splits are scored once.

4.1 SETUP

Datasets. **DROID** (Khazatsky et al., 2024): 1,126 real Franka episodes from 24 official shards, split by recording session (851/143/132 train/val/test). We use the primary exterior camera and report the second exterior camera as a transfer view. One model step is 5 frames (0.33 s) with a 35-D block of Cartesian and gripper commands, and we forecast up to $K=10$ steps (3.3 s). **Open-H-Embodiment, Hamlyn dVRK** (Open-H-Embodiment Consortium, 2026): seven surgical tasks (suturing $\times 2$, knot tying, needle grasp and handover, peg transfer, tissue lifting, tissue retraction) with endoscope video and recorded robot kinematics as actions. Splits are episode-disjoint per task (70/15/15). **IWS** (Wang et al., 2026; Zhang et al., 2026a): single-image forecasting on PushT, Box and Rope with the official RLA-WM evaluation handles. **Planning:** PushT, TwoRoom and Reacher from the LeWM/stable-worldmodel suite (Maes et al., 2026b;a), with official planner budgets.

Encoders and models. Frozen DINOv2-S/14 and DINOv2-B/14 (Oquab et al., 2024) (16×16 patches at 224 px) and DINOv3-S/16 (Siméoni et al., 2025) (16×16 at 256 px). No pooling is applied. All learned predictors use width 384 with $L_e=L_d=4$ ($\approx$ pendingM parameters) and the same optimiser, schedule, number of steps and batch size (Appendix A).

Metrics. Forecasting: standardised feature MSE per horizon (mean over horizons and at the final horizon), cosine similarity, and MSE after 4×4 pooling for comparison with prior work. Decoded-pixel: PSNR, SSIM and LPIPS through a shared feature-to-RGB decoder. Physical grounding: linear probes of end-effector or kinematic state from predicted features. Action sensitivity: error under shuffled actions and the rate at which true actions rank above shuffled ones. Planning: success rate and planning time. Statistics: episodes are the unit of analysis, and we report paired bootstrap 95% intervals over episodes, pooled across seeds.

270 4.2 HELD-OUT FORECASTING ON REAL ROBOT AND SURGICAL VIDEO
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272 4.3 ENCODERS, HORIZONS AND COMPONENTS
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274 pending
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276 4.4 PLANNING
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278 pending
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280 5 ANALYSIS
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282 **What does the transport field learn?** pending
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284 **Where do the gains come from?** pending
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286 **Accuracy versus action sensitivity.** pending
287
288 **Efficiency.** pending
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290 6 CONCLUSION
291
292 We presented SHIFTWM, a latent world model that forecasts by transporting observed foundation
293 features and synthesising only what transport cannot explain. pending
294
295 **Limitations.** SHIFTWM predicts features, not pixels. Visualisations and pixel metrics rely on a
296 separately trained decoder. Transport copies only content that has been observed, so newly revealed
297 regions depend on the correction term. The action-contrastive objective uses in-batch negatives,
298 which may be easy to distinguish in diverse datasets.
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300 REPRODUCIBILITY STATEMENT
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302 Code, configurations, per-episode evaluation records and trained checkpoints will be released. Ap-
303 pendix A lists every hyperparameter. Dataset splits are deterministic and documented in Ap-
304 pendix B. All datasets are public: DROID (CC-BY-4.0), Open-H-Embodiment (CC-BY-4.0), the
305 RLA-WM/IWS release, and the LeWM planning datasets (MIT).
306
307 ETHICS STATEMENT
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309 The surgical data are public recordings of robotic procedures on phantoms and ex-vivo tissue, re-
310 leased under CC-BY-4.0. We make no clinical claims, and the models are not intended for clinical
311 use.
312
313 AI USE STATEMENT
314
315 In this work, we used generative AI tools to assist with writing and debugging research code, or-
316 chestrating experiments, surveying related work, and editing the manuscript text. All AI-assisted
317 code was tested, and every reported number is produced by logged runs that the authors reviewed.
318 Bibliographic entries were verified against the original sources. We take responsibility for the final
319 content of this work, including text, claims and artifacts produced with the aid of generative AI.
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